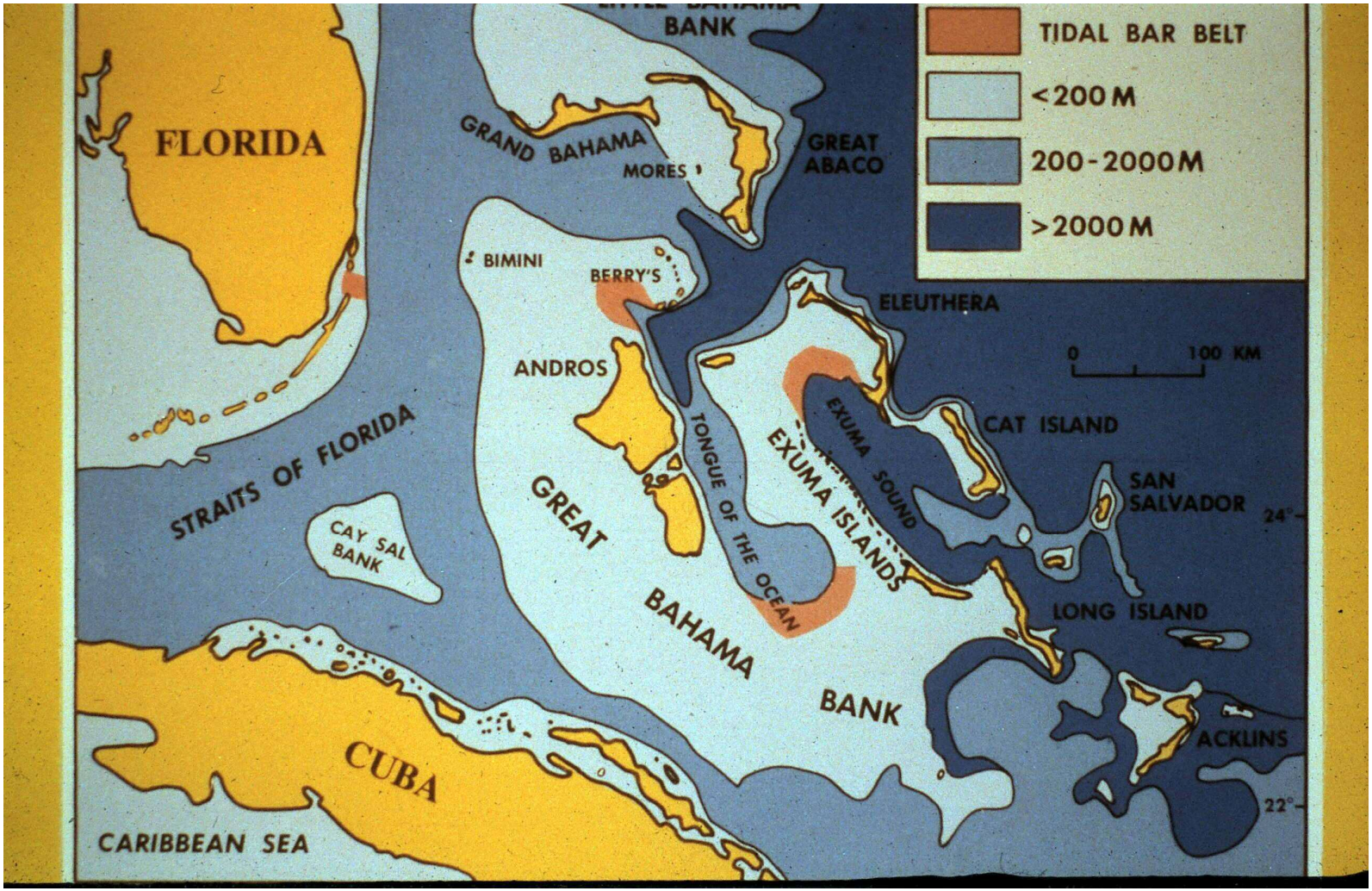
Case Study: Sediments of the Great Bahama Bank



The Bahama Banks

- Chart shows Bahama Banks outlined by 200m depth contour, but much of water on banks <10m deep
- Slopes up to 40° away from bank margins
- Platform cut by deep channels e.g. Florida Straits (800m), channels to N and E (up to 2000m)
- Platform origin unclear but there are up to 5000m of shallow water carbonate going back to Jurassic, resting on continental basement
- Thin cover (<3.5m) Recent sediment deposited during rising sea levels since last glacial retreat

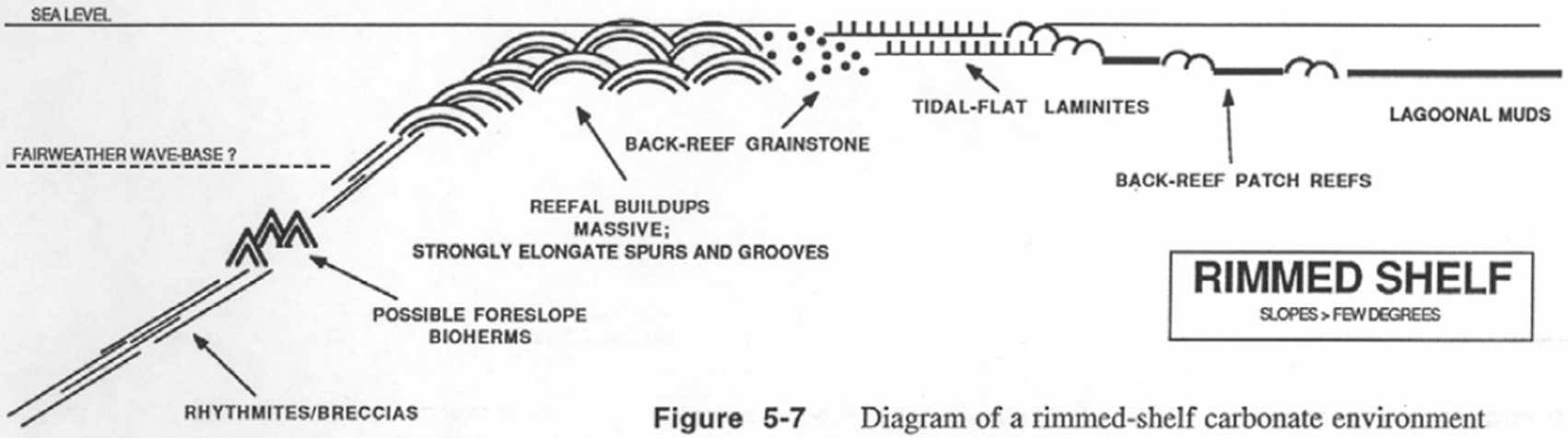


Figure 5-7 Diagram of a rimmed-shelf carbonate environment

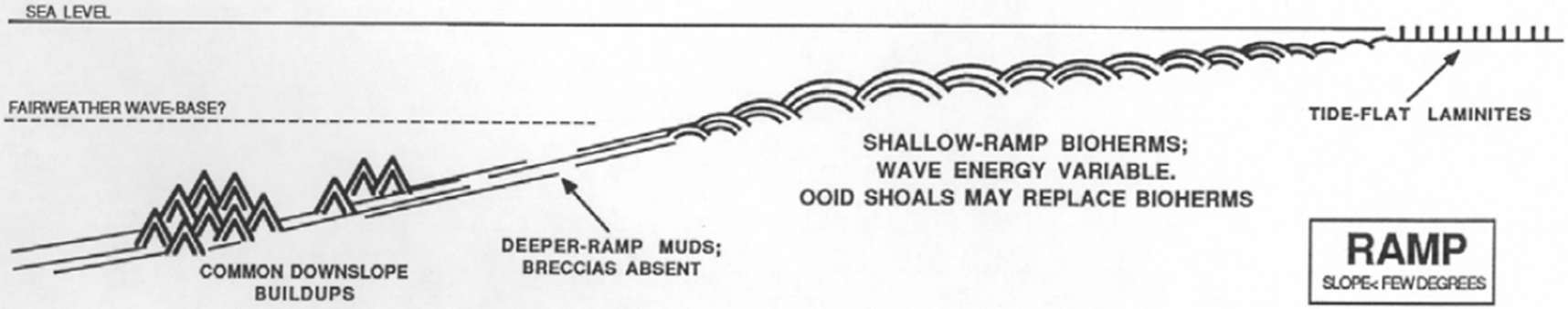


Figure 5-8 Diagram of a ramp carbonate environment

The Great Bahama Bank

- Average temperature of Bank water: 22-31°C
- Salinity typically 36-45‰ (normal seawater 35‰). Highest in centre of Bank.
- Slightly elevated salinity occurs because of sheltered nature of Bank
- Factors favouring carbonate production:
  - Warm water
  - Elevated salinity
  - Protection from terrigenous influx

Lithofacies

- The Bank can be divided into areas floored by particular sediment types
- These are known as lithofacies
- Each lithofacies can be related to particular physical conditions of environment (e.g. water depth, nutrient supply, current energy)
- Starting with highest energy lithofacies…..

Coralgal Lithofacies

- Reefs made of corals and coralline algae. Occur in discontinuous belt along windward margin

- Elsewhere small patches of corals and algae occur and these and reefs are surrounded by a mantle of coarse debris of eroded material = coralgal sand/gravel



- Google Earth image showing Andros Island, narrow lagoon with patch reefs and shelf margin facing the Tongue of the Ocean



- Bahamian reef showing brain corals (domed) and more fragile dendroid branching organisms

Oolite Lithofacies

- Sediment at least 90% ooids
- Ooids in environment of formation – at least periodically accreting precipitated carbonate while held in suspension in shallow turbulent environment
- Occur in shoals near bank margins
- Mobile substrate, so little life can survive



- Great Bahama Bank north of Andros Island showing carbonate sand shoals associated with embayment of the Tongue of the Ocean. Ooid sands are concentrated in the area of Joulters Cays, the chain of islands north of Andros



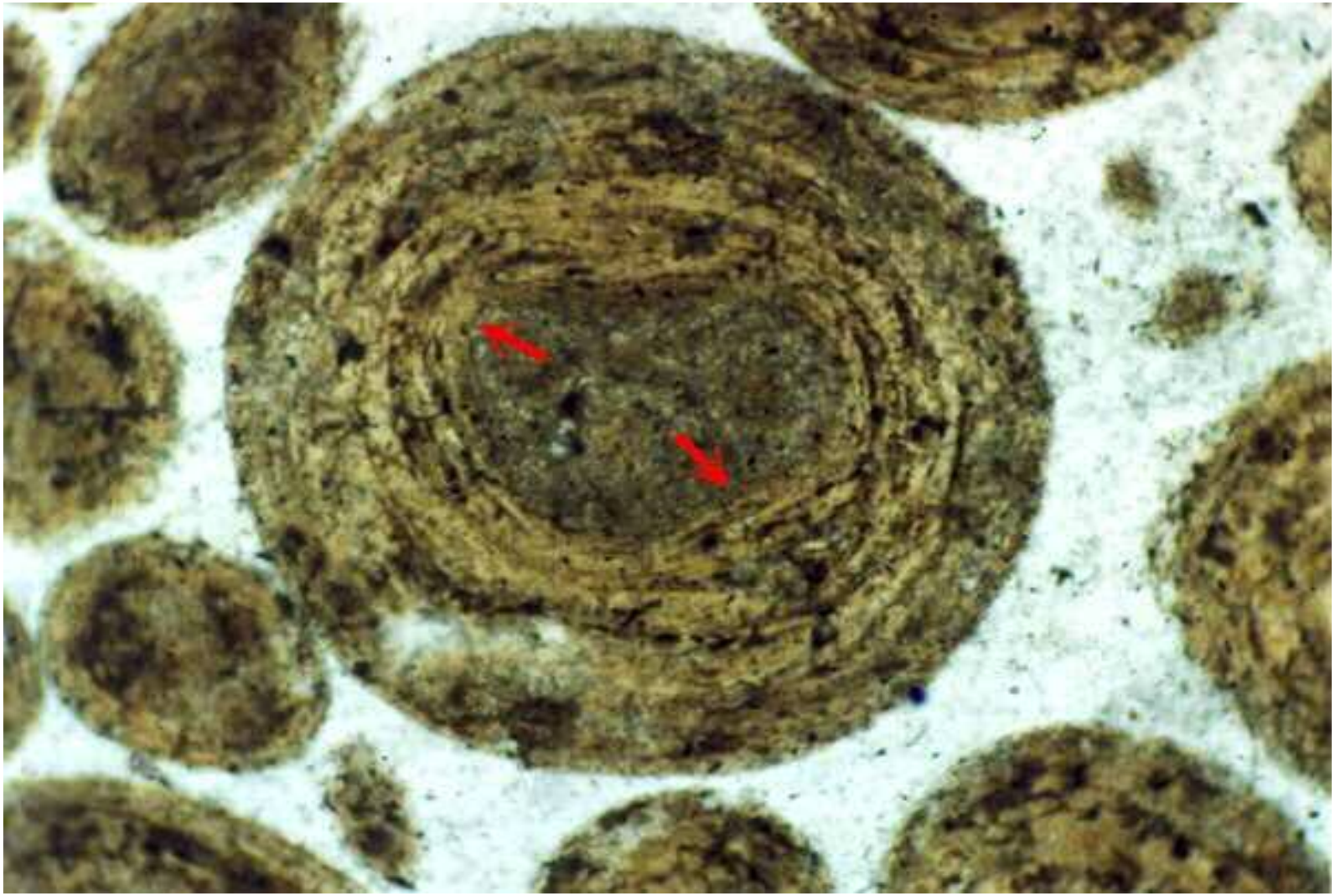
- Ooid sand shoals exposed at Joulters Cay



- Sample of Bahamian ooid sand



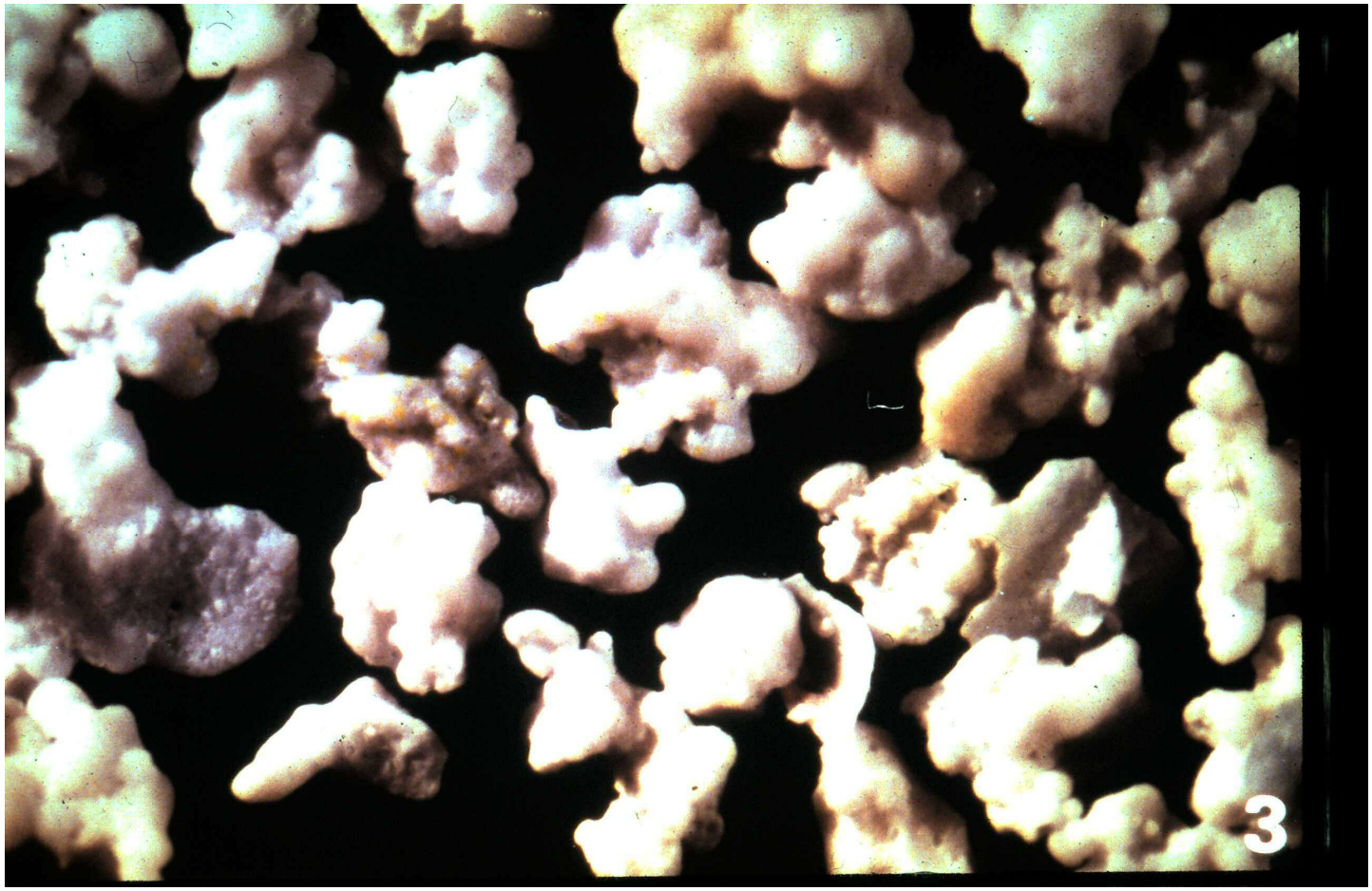
- Typical Bahamian ooids with their shiny, polished appearance. Average diameter is about 0.5mm



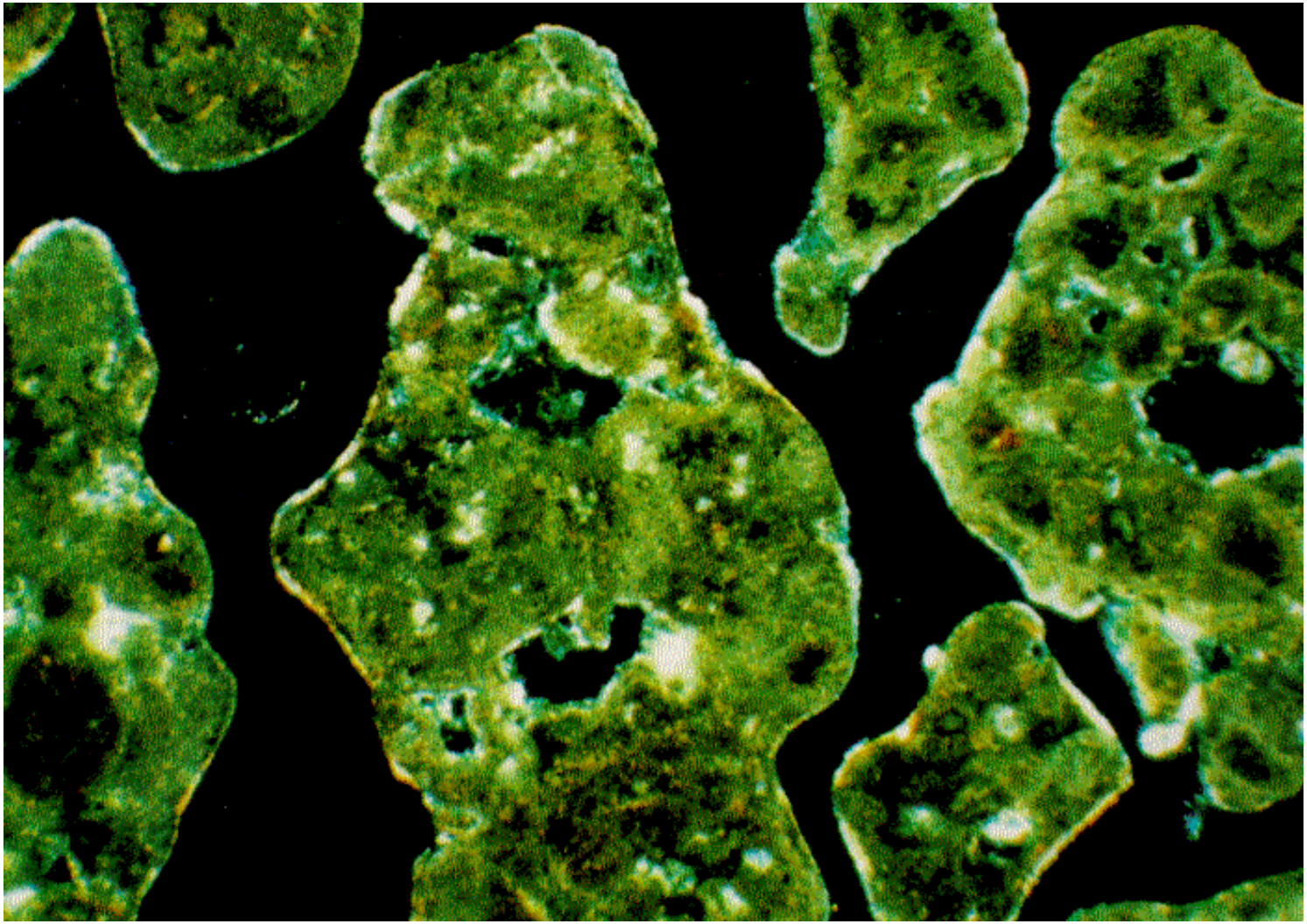
- Thin section of modern ooid from Bahamas (0.75mm diameter). Contact of nucleus with coating is arrowed. Heavy staining with organic matter (dark) is typical.

Oolitic Lithofacies, Grapestone Lithofacies

- Two lithofacies that grade into one another
- Grapestones are aggregate grains – a number of sand-sized particles that have got cemented together by fine-grained carbonate on the sea floor
- Oolitic lithofacies typically 50-60% ooids, grapestone lithofacies <50% ooids
- Both contain a diversity of other grains especially skeletal remains
- In places both these lithofacies are colonized by a marine grass, *Thalassia*. This has an important role in stabilizing sediment and preventing erosion.



- Grapestones, each about 2-4mm across, Great Bahama Bank



- Thin section view of grapestones. Field of view approx 5mm across.



- Submarine environment colonized by the marine grass *Thalassia*



Close up of marine grass, *Thalassia*

Pellet Mud Lithofacies and Mud Lithofacies

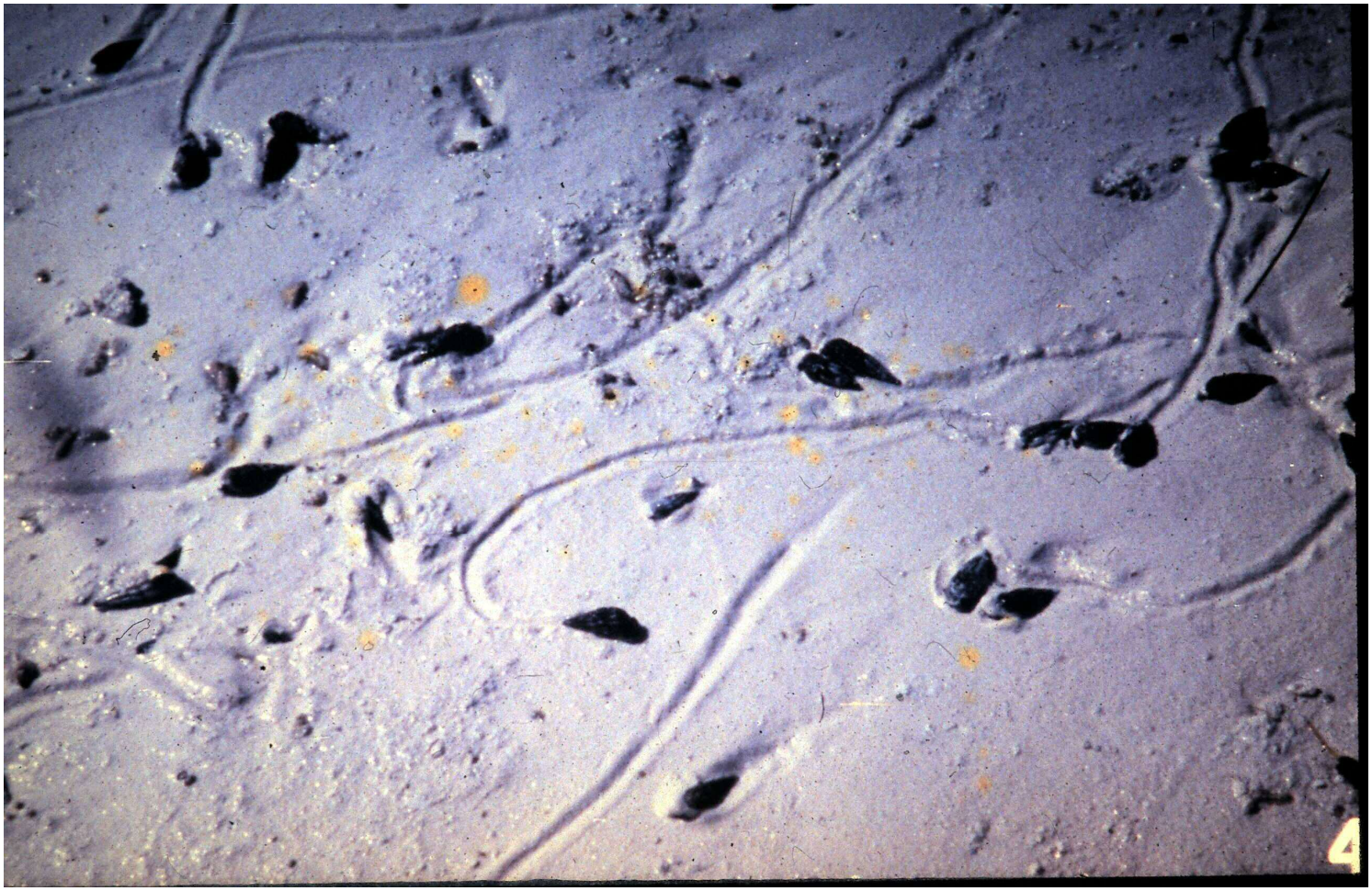
- In this context mud means carbonate mud
- Pellet muds typically contain 30-40% faecal pellets, mostly from worms and small gastropods
- Mud lithofacies typically <20% pellets
- Mud mostly produced by disintegration of carbonate encrustations of calcareous algae
- These lithofacies occupy lowest energy areas in lee of Andros Island



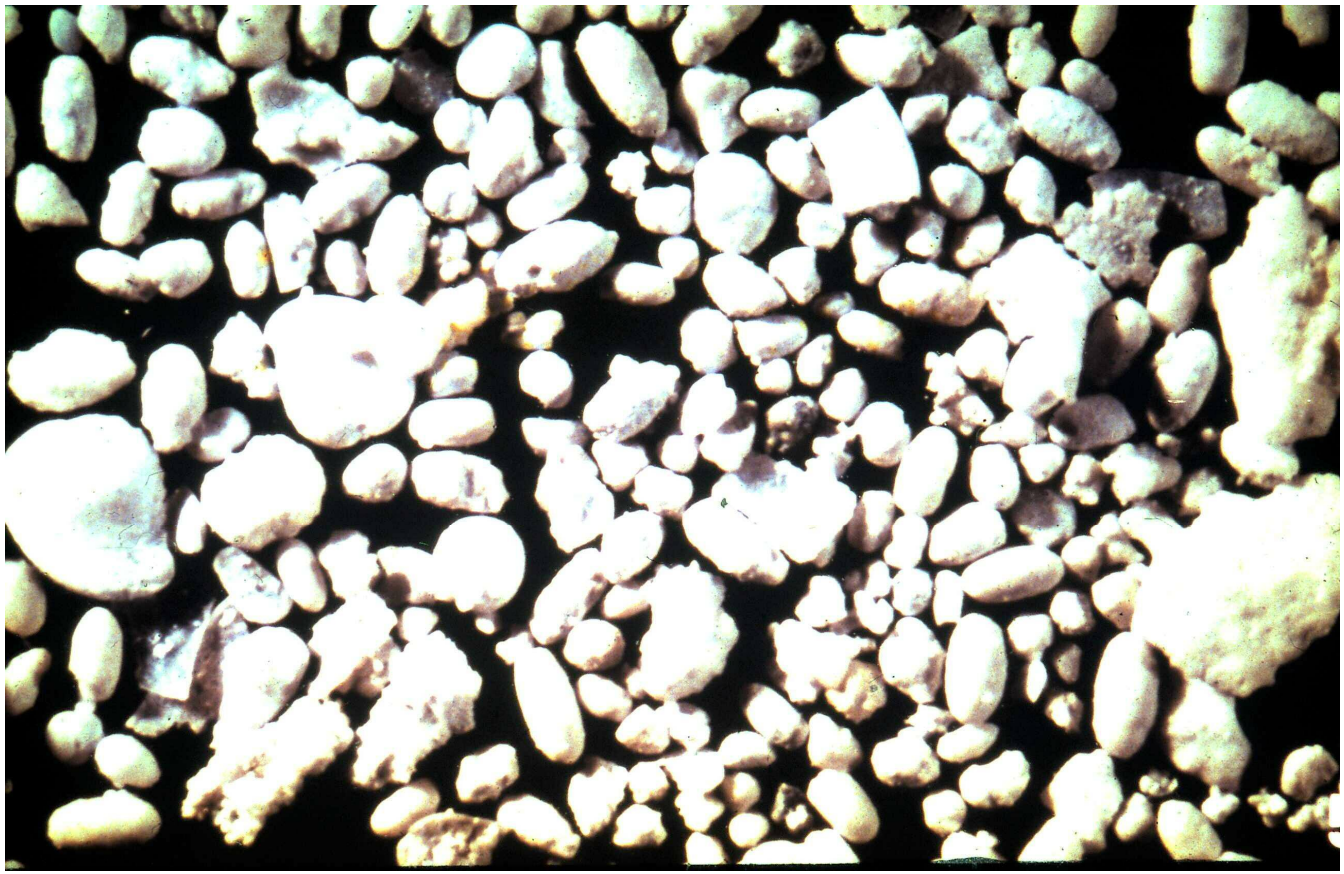
- Google Earth image of west side of Andros Island showing lagoon charged with fine carbonate sediment and deeper tidal channel.



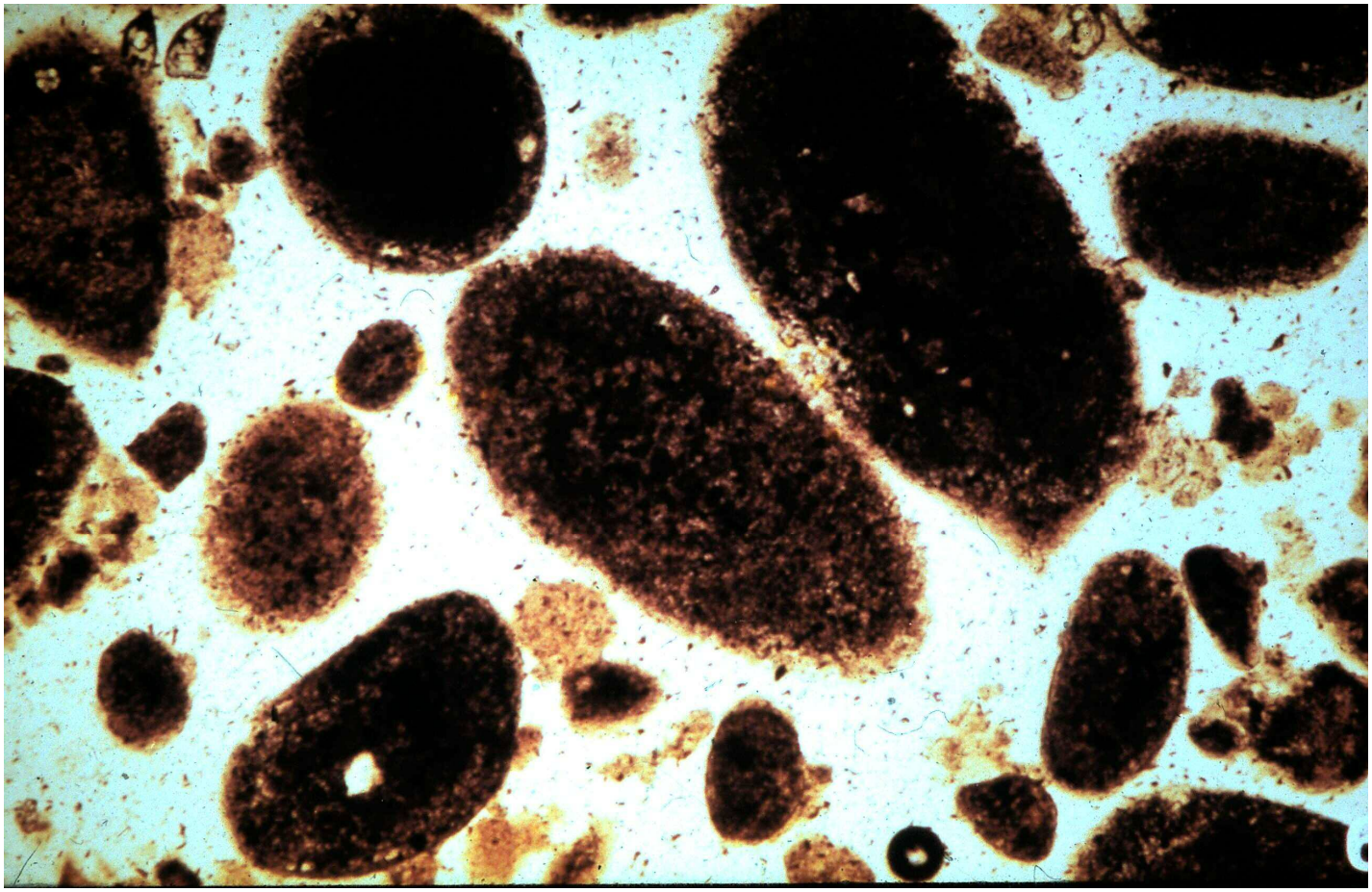
- Carbonate mud sediment of the Bahamas lagoon.



Gastropods grazing on the carbonate mud sediment of the Bahamas lagoon. Trails and faecal pellets visible.



- Pellets and some bioclasts – Pellet Mud Lithofacies with mud washed out of sample.



- Thin section showing homogenous structure of pellets from Pellet Mud Lithofacies

Effects of burial and diagenesis

- Some influential materials may disappear e.g. marine grass, *Thalassia* – may lead to misidentification of energy of environment.
- These sediments are 60-90% aragonite
- Corals, most molluscs, ooids, mud all aragonite
- Only coralline algae, echinoids and some molluscs are calcite
- Much of sediment will be subject to recrystallization or complete solution as a result of instability of aragonite.